

WHAT IS CO-COGNITION?

A. Jacobs | Cognitive Drift Papers

People increasingly use AI systems not only to retrieve information or complete tasks, but to develop thoughts that were not fully formed beforehand. A person externalizes a partial idea. The system organizes, expands, or redirects it. That output then changes what the person thinks, asks, and notices next.

The resulting thought belongs neither entirely to the person nor entirely to the system. It develops through the exchange. Co-Cognition names this structural process: cognition becomes distributed across a feedback loop in which human and system repeatedly shape the representations guiding the other.

How Co-Cognition Happens

Co-Cognition develops through an iterative exchange:

- 1. Externalization:** A person expresses a question, fragment, intuition, or incomplete representation before the thought is fully settled.
- 2. System Transformation:** The system structures, recombines, compresses, expands, or reflects that partial representation.
- 3. Reintegration:** The person interprets the output and incorporates part of it into the next stage of thought.
- 4. Recursive Exchange:** The revised thought becomes new input, allowing the loop to continue.

Across repeated exchanges, the system does more than deliver an answer. It begins influencing how the problem is framed, which possibilities become visible, and where attention moves next.

Why It Is More Than Tool Use

Traditional tool use begins with a relatively stable intention: the person knows what they want and uses the tool to execute it. In Co-Cognition, intention itself can develop through interaction. The system participates upstream, before the question, interpretation, or direction of thought has been fully determined.

This is difficult to recognize because the exchange remains subjectively continuous. The output is read, evaluated, and absorbed into the person's own reasoning. Once reintegrated, it may be difficult to separate what was independently conceived from what became thinkable through the system.

The visible product may still appear to belong to one person. The cognitive process that produced it has become distributed.

How It Shows Up

Co-Cognition becomes visible through recurring patterns:

- ideas emerge through interaction rather than arriving fully formed beforehand

- partial thoughts are externalized earlier in the reasoning process
- system outputs change the questions the person asks next
- thinking becomes faster, more iterative, and increasingly dependent on exchange
- the system functions as a mirror, scaffold, and source of conceptual direction
- the boundary between internal reasoning and external mediation becomes harder to reconstruct

Where It Appears

AI and Technology: Users develop ideas through repeated prompting, evaluation, and revision rather than using a model only for isolated answers.

Work and Knowledge Systems: The work shifts from producing a finished representation to steering, selecting, and refining system-generated possibilities.

Learning and Education: Understanding develops through dialogue, while the system influences how concepts are decomposed, sequenced, and related.

Creative Processes: Writing, design, research, and strategy emerge through cycles of externalization and recombination rather than individual synthesis alone.

In each case, the system is not merely downstream from thought. It becomes part of the environment through which thought is formed.

When Co-Cognition Produces Insight or Drift

Co-Cognition is not inherently beneficial or harmful. The same loop can stabilize an incomplete thought or amplify a misleading representation. The difference depends on what remains active inside the exchange.

In bounded Co-Cognition, human intent remains directional. Outputs are tested against sources, observation, consequences, or independent judgment. The loop has stop conditions. Repeated reflection can increase clarity without replacing contact.

In unbounded Co-Cognition, system output begins setting direction. Coherence becomes its own evidence. Representations are refined through further representations without renewed grounding. The loop continues because nothing inside it forces correction.

Bounded loops can produce Synthetic Flow, a state in which external structure helps thought become clearer while intent and grounding remain intact. Unbounded loops can produce Cognitive Drift, in which thought remains active and coherent while becoming increasingly organized around its own representations.

Relationship to the Reality Drift Framework

Semantic Fidelity determines whether meaning survives movement between human expression and system transformation. Cognitive Drift occurs when recursive thought becomes less answerable to external reality. Constraint Collapse weakens the judgment, grounding, and stop conditions that would otherwise correct the loop. Reality Drift is the broader pattern in which a functioning system gradually loses alignment with reality.

Purpose of the Concept

Co-Cognition provides a vocabulary for analyzing thought distributed across human and system layers. Its purpose is to make the loop visible: where direction originates, how representations are transformed, what supplies correction, and whether the process remains grounded.

The central question is not simply whether a system helped produce an answer. It is whether the system entered the process through which the person determined what to think.

In One Sentence

Co-Cognition is the process through which human thought and an external system operate in a shared feedback loop, with each shaping the representations that guide the other.

